MSG No. 123.340 $P4/mmm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `1a`, site symmetry: `4/mmm1'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24, 25,26,27,28,29,30,31,32]

Table 2: Wyckoff site: `1b`, site symmetry: `4/mmm1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24, 25,26,27,28,29,30,31,32]

Table 3: Wyckoff site: `1c`, site symmetry: `4/mmm1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24, 25,26,27,28,29,30,31,32]

Table 4: Wyckoff site: `1d`, site symmetry: `4/mmm1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24, 25,26,27,28,29,30,31,32]

Table 5: Wyckoff site: `2e`, site symmetry: `mmm.1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 6: Wyckoff site: `2f`, site symmetry: `mmm.1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 7: Wyckoff site: `2g`, site symmetry: `4mm1'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,6,12,13,15,16,17,18,19,22,28,29,31,32]

2	$[0,\ 0,\ -z]$	[4,5,7,8,9,10,11,14,20,21,23,24,25,26,27,30]

Table 8: Wyckoff site: 2h, site symmetry: 4mm1'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[1,2,3,6,12,13,15,16,17,18,19,22,28,29,31,32]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[4,5,7,8,9,10,11,14,20,21,23,24,25,26,27,30]

Table 9: Wyckoff site: 4i, site symmetry: 2mm.1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,6,12,13,17,22,28,29]
2	$\left[\frac{1}{2},\ 0,\ z\right]$	[2,3,15,16,18,19,31,32]
3	$\left[0,\ \frac{1}{2},\ -z\right]$	[4,5,9,14,20,21,25,30]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[7,8,10,11,23,24,26,27]

Table 10: Wyckoff site: 4j, site symmetry: m.2m1'

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,7,14,16,17,23,30,32]
2	$[-x,\ x,\ 0]$	[2,5,11,12,18,21,27,28]
3	$[x,\ -x,\ 0]$	[3,4,10,13,19,20,26,29]
4	$[-x,\ -x,\ 0]$	[6,8,9,15,22,24,25,31]

Table 11: Wyckoff site: 4k, site symmetry: m.2m1'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	[1,7,14,16,17,23,30,32]
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	[2,5,11,12,18,21,27,28]
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	[3,4,10,13,19,20,26,29]
4	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[6,8,9,15,22,24,25,31]

Table 12: Wyckoff site: 4l, site symmetry: m2m.1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4,13,14,17,20,29,30]
2	$[0,\ x,\ 0]$	[2,7,11,16,18,23,27,32]
3	$[0,\ -x,\ 0]$	[3,8,10,15,19,24,26,31]
4	$[-x,\ 0,\ 0]$	[5,6,9,12,21,22,25,28]

Table 13: Wyckoff site: 4m, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,4,13,14,17,20,29,30]
2	$\left[0,\ x,\ \frac{1}{2}\right]$	[2,7,11,16,18,23,27,32]
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	[3,8,10,15,19,24,26,31]
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[5,6,9,12,21,22,25,28]

Table 14: Wyckoff site: 4n, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,4,13,14,17,20,29,30]
2	$\left[\frac{1}{2},\ x,\ 0\right]$	[2,7,11,16,18,23,27,32]
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	[3,8,10,15,19,24,26,31]
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,6,9,12,21,22,25,28]

Table 15: Wyckoff site: 4o, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,4,13,14,17,20,29,30]
2	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[2,7,11,16,18,23,27,32]
3	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[3,8,10,15,19,24,26,31]
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[5,6,9,12,21,22,25,28]

Table 16: Wyckoff site: 8p, site symmetry: m..1'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14,17,30]
2	$[-y,\ x,\ 0]$	[2,11,18,27]
3	$[y,\ -x,\ 0]$	[3,10,19,26]
4	$[x,\ -y,\ 0]$	[4,13,20,29]
5	$[-x,\ y,\ 0]$	[5,12,21,28]
6	$[-x,\ -y,\ 0]$	[6,9,22,25]
7	$[y,\ x,\ 0]$	[7,16,23,32]
8	$[-y,\ -x,\ 0]$	[8,15,24,31]

Table 17: Wyckoff site: 8q, site symmetry: m..1'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,14,17,30]
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	[2,11,18,27]
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	[3,10,19,26]
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	[4,13,20,29]
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	[5,12,21,28]
6	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[6,9,22,25]

continued ...

Table 17

No.	position	mapping
7	$\left[y,\, x,\, \frac{1}{2}\right]$	[7,16,23,32]
8	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[8,15,24,31]

Table 18: Wyckoff site: 8r, site symmetry: ..m1'

No.	position	mapping
1	$[x,\, x,\, z]$	[1,16,17,32]
2	$[-x,\, x,\, z]$	[2,12,18,28]
3	$[x,\, -x,\, z]$	[3,13,19,29]
4	$[x,\, -x,\, -z]$	[4,10,20,26]
5	$[-x,\, x,\, -z]$	[5,11,21,27]
6	$[-x,\, -x,\, z]$	[6,15,22,31]
7	$[x,\, x,\, -z]$	[7,14,23,30]
8	$[-x,\, -x,\, -z]$	[8,9,24,25]

Table 19: Wyckoff site: 8s, site symmetry: .m.1'

No.	position	mapping
1	$[x,\, 0,\, z]$	[1,13,17,29]
2	$[0,\, x,\, z]$	[2,16,18,32]
3	$[0,\, -x,\, z]$	[3,15,19,31]
4	$[x,\, 0,\, -z]$	[4,14,20,30]
5	$[-x,\, 0,\, -z]$	[5,9,21,25]
6	$[-x,\, 0,\, z]$	[6,12,22,28]
7	$[0,\, x,\, -z]$	[7,11,23,27]
8	$[0,\, -x,\, -z]$	[8,10,24,26]

Table 20: Wyckoff site: 8t, site symmetry: .m.1'

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, z\right]$	[1,13,17,29]
2	$\left[\frac{1}{2},\, x,\, z\right]$	[2,16,18,32]
3	$\left[\frac{1}{2},\, -x,\, z\right]$	[3,15,19,31]
4	$\left[x,\, \frac{1}{2},\, -z\right]$	[4,14,20,30]
5	$\left[-x,\, \frac{1}{2},\, -z\right]$	[5,9,21,25]
6	$\left[-x,\, \frac{1}{2},\, z\right]$	[6,12,22,28]
7	$\left[\frac{1}{2},\, x,\, -z\right]$	[7,11,23,27]
8	$\left[\frac{1}{2},\, -x,\, -z\right]$	[8,10,24,26]

Table 21: Wyckoff site: 16u, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,17]
2	$[-y, x, z]$	[2,18]
3	$[y, -x, z]$	[3,19]
4	$[x, -y, -z]$	[4,20]
5	$[-x, y, -z]$	[5,21]
6	$[-x, -y, z]$	[6,22]
7	$[y, x, -z]$	[7,23]
8	$[-y, -x, -z]$	[8,24]
9	$[-x, -y, -z]$	[9,25]
10	$[y, -x, -z]$	[10,26]
11	$[-y, x, -z]$	[11,27]
12	$[-x, y, z]$	[12,28]
13	$[x, -y, z]$	[13,29]
14	$[x, y, -z]$	[14,30]
15	$[-y, -x, z]$	[15,31]
16	$[y, x, z]$	[16,32]